

Section: Particles and Waves

Topic: The Standard Model

Question 7 – The Matryoshka Analogy

✓ 3 marks

A student says:

“Particle physics is just like matryoshka dolls. Look inside a particle and you will always find something similar, just smaller.”

Using your knowledge of physics, comment on this statement.

✓ Full-mark Response (3/3)

The **Standard Model** of particle physics tells us that **fundamental particles** (such as quarks, electrons, and neutrinos) have **no internal structure**.

Unlike matryoshka dolls, where each doll contains a smaller one, **fundamental particles are not made up of smaller parts**.

For example, **protons and neutrons are composite** particles made of **quarks**, but quarks themselves are thought to be **elementary**.

Therefore, the analogy breaks down at the fundamental level — you **do not always find something smaller** inside.

Revision Tips

- The **Standard Model** contains both **composite** and **elementary** particles
 - Composite: **protons, neutrons** → made of **quarks**
 - Elementary: **quarks, electrons, neutrinos, photons** → no smaller parts
- Particle physics seeks the most **basic building blocks** — it **does not go on forever**